COMPSCI 683: Artificial Intelligence

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Local Search

AIMA Chapter 4.1

Local Search Algorithms

- The path to goal is irrelevant; the goal state itself is the solution
- State space = set of “complete” configurations
- Find final configuration satisfying constraints
- Local search algorithms: maintain single “current best” state and try to improve it
- Advantages:
  - very little/constant memory
  - find reasonable solutions in large state space

Local Search Algorithms

Boolean satisfiability: given a CNF Boolean formula $\phi(x_1, \ldots, x_n)$, find a satisfying truth assignment.

Start with some arbitrary truth assignment (e.g. all variables are initially $False$), change truth values one variable at a time until a satisfying assignment has been found.

Local Search Algorithms

Gradient Descent: given a differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, find a minimum of f.

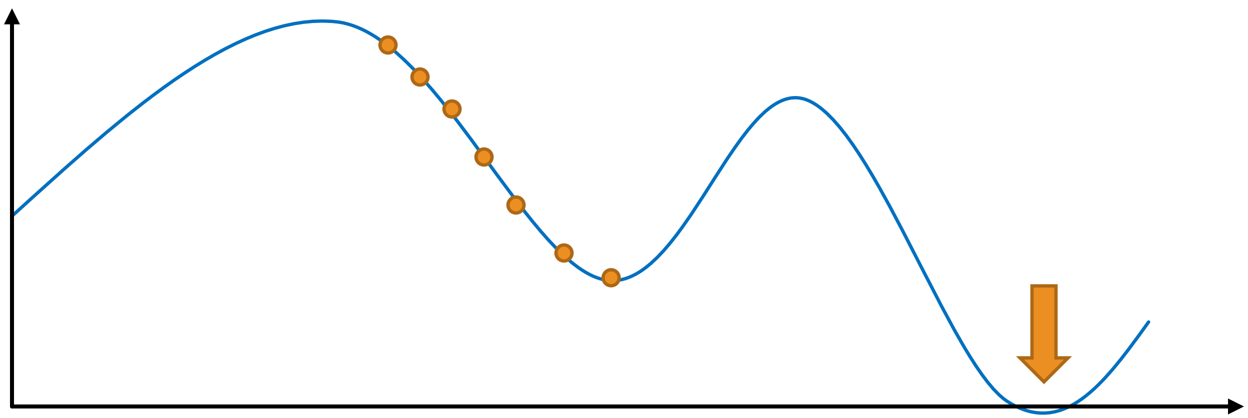
Start from some $\vec{x}^0$. At every round, set

$$\vec{x}_i^t = \vec{x}_i^{t-1} - \eta \times \frac{\partial f}{\partial x_i}(\vec{x}^{t-1})$$

Gradient of f at $\vec{x}^{t-1}$

Step size

Stop once $f(\vec{x}^{t-1}) \leq f(\vec{x}^t)$.



Example: n-queens

- Put n queens on an $n \times n$ board with no two queens on the same row, column, or diagonal

Hill-Climbing Search

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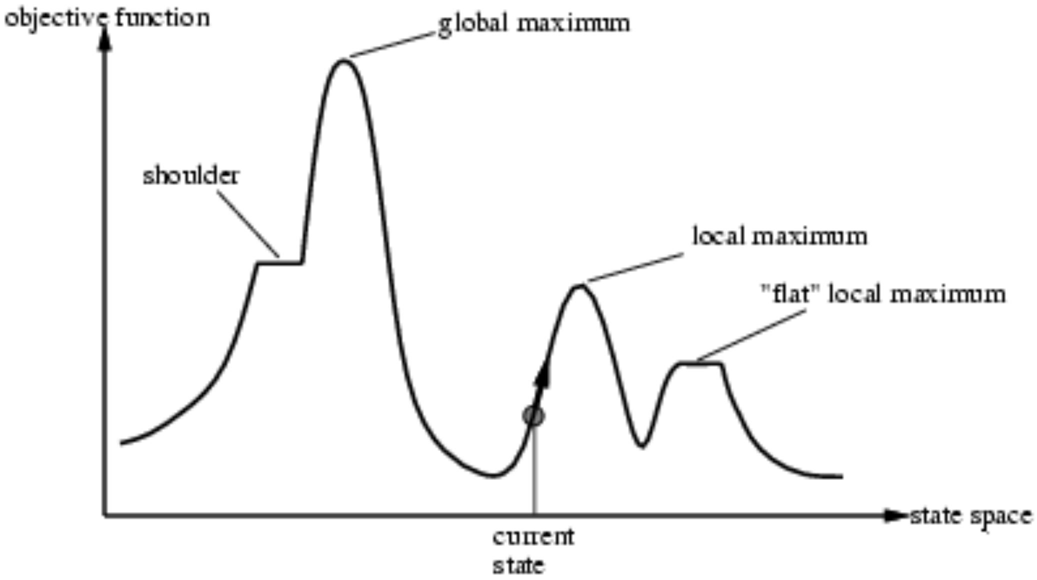
function HILL-CLIMBING(problem) returns a state that is a local maximum

  current ← MAKE-NODE(problem.INITIAL-STATE)
  loop do
      neighbor ← a highest-valued successor of current
      if neighbor.VALUE ≤ current.VALUE then return current.STATE
      current ← neighbor
```

"Like climbing Mt. Everest in thick fog with amnesia"

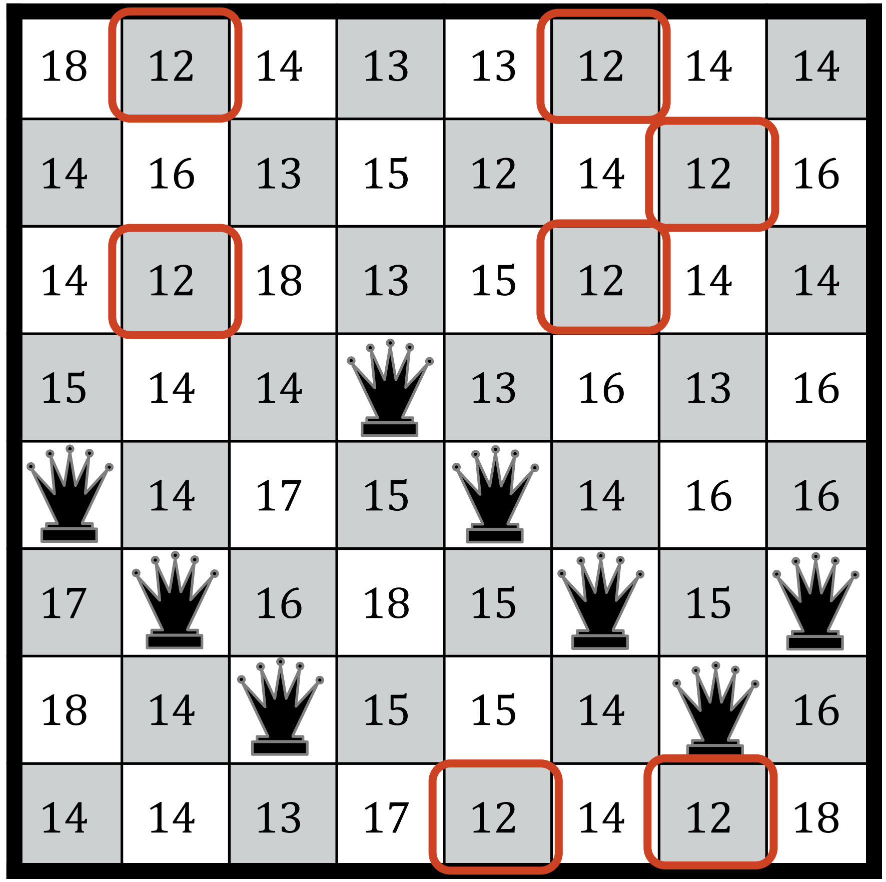
Hill-Climbing Search

- Problem: depending on initial state, can get stuck in local maxima



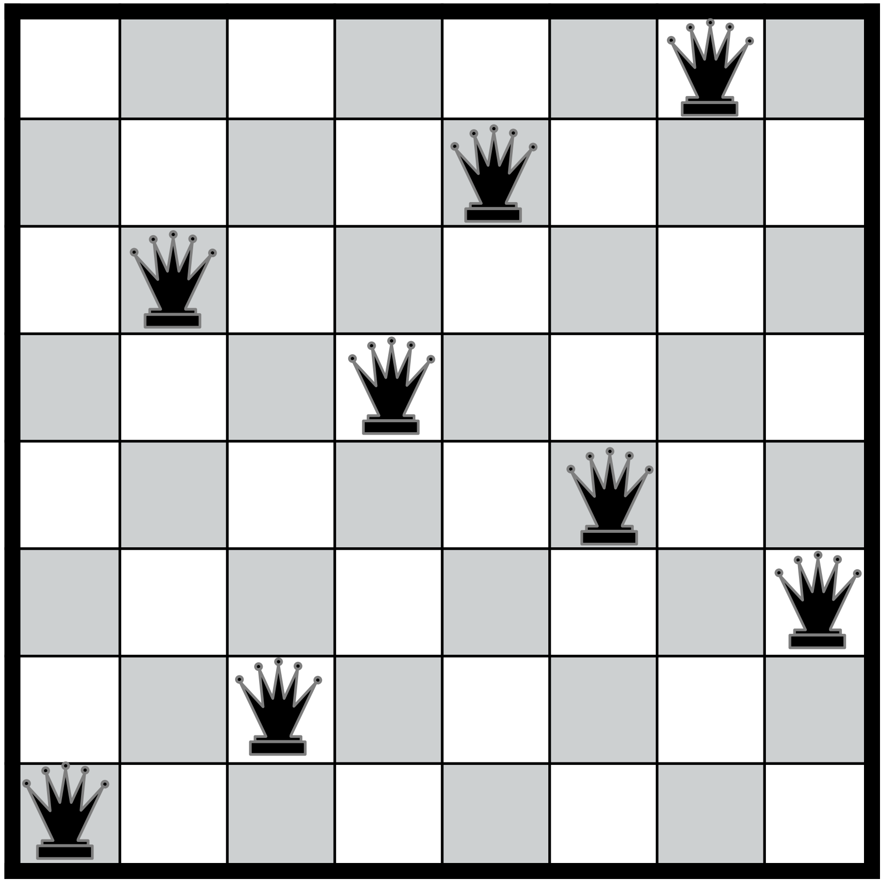
- Non-guaranteed fixes: sideway moves, random restarts

Hill-Climbing Search: 8-Queens



- $h =$ number of pairs of queens that are attacking each other, either directly or indirectly
- $h = 17$ for the above state

Hill-Climbing Search: 8-Queens



Local Minimum with $h = 1$